ECOVISION ACADEMY

PYTHON FOR DATA SCIENCE

Foundation Course in Python for Data Science (25 Hours)

Course Fee:	Total Duration:	Level:
₹399 Only	25 Hours	Beginner to Advanced

✓ **Outcome:** Strong foundation for Data Analysis & Data Science.

python

DATA SCIENCE

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PYTHON FOR DATA SCIENCE COURSE

SECTION 1: Introduction to Data Science & Python Environment (2 Hours)

- What is Data Science?
- Data vs Information vs Knowledge
- Data Science lifecycle (Collect → Clean → Analyze → Visualize → Decide)
- Roles in Data Field: Data Analyst, Data Scientist, ML Engineer
- Types of Data: Structured, Semi-structured, Unstructured
- Python Ecosystem for Data Science
- Python Installation & Environment Setup (Anaconda, Jupyter Notebook)

SECTION 2: Python Basics – Core Programming Logic (4 Hours)

- Python Syntax & Code Structure
- Variables, Data Types & Type Casting
- Operators (Arithmetic, Logical, Comparison)
- Input & Output Operations
- Conditional Statements (if, elif, else)
- Looping Concepts (for, while)
- Flow Control (break, continue, pass)
- Writing Clean & Readable Code

SECTION 3: Python Data Structures (4 Hours)

- Strings: Indexing, slicing, string methods
- Lists: Creation, updating, list methods
- Tuples & Sets: Difference, immutability, use cases
- Dictionaries: Key-value mapping, nested dictionaries
- Data Structure Selection
- Iterating through Data Structures
- Practice Problems & Mini Exercises

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SECTION 4: NumPy for Numerical & Scientific Computing (4 Hours)

- Introduction to NumPy & ndarray
- Creating Arrays (1D, 2D, 3D)
- Indexing, Slicing & Boolean Indexing
- Array Operations & Broadcasting
- Mathematical Functions (Mean, Median, Standard Deviation)
- Random Module & Sampling
- Performance Advantage of NumPy over Python Lists
- Hands-on Practice with Numeric Dataset

SECTION 5: Pandas for Data Analysis (4 Hours)

- Introduction to Pandas: Series vs DataFrame
- Importing Data (CSV, Excel)
- Exploring Data (head, tail, info, describe)
- Data Cleaning: Missing values, duplicates
- Data Selection using loc & iloc
- Sorting, GroupBy & Aggregation
- Basic Feature Engineering Concepts
- Real-world Dataset Case Study

SECTION 6: Data Visualization – Matplotlib & Seaborn (4 Hours)

- Importance of Data Visualization
- Types of Charts & Selection
- Matplotlib: Line, Bar, Histogram, Scatter plots
- Customizing Plots (labels, titles, legends)
- Seaborn Introduction & Themes
- Statistical Plots: Boxplot, Heatmap, Distribution
- Visualization Best Practices
- Graph Interpretation & Insights

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SECTION 7: Introduction to Machine Learning (1 Hour)

- What is Machine Learning?
- Types: Supervised & Unsupervised
- Machine Learning Workflow
- Linear Regression Intuition
- Overfitting & Underfitting

SECTION 8: Mini Project, Assessment & Certification (2 Hours)

- End-to-End Mini Project
- Data Cleaning, Analysis & Visualization
- Project Presentation & Discussion
- Foundation-level Interview Questions,
- Career Guidance & Certification Process